Law of Sines and Law of Cosines

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Law of Sines**

In triangle ABC, $A = 40°$, $B = 65°$, and $a = 12$. Find:

a angle C $= 75°$

b side b (2 d.p.) $= \frac{12\sin 65°}{\sin 40°} \approx 16.92$

c side c (2 d.p.) $= \frac{12\sin 75°}{\sin 40°} \approx 18.03$

(a) $= 75°$; (b) $= \frac{12\sin 65°}{\sin 40°} \approx 16.92$; (c) $= \frac{12\sin 75°}{\sin 40°} \approx 18.03$

2 **[Foundational]** • **Target: Law of Sines**

Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are $58°$ and $47°$. Find the distance from the balloon to the first observer (the one with the $58°$ sighting), to the nearest metre.

The angle at the balloon is $180° - 58° - 47° = 75°$. By the Law of Sines, $d = \frac{400\sin 47°}{\sin 75°} \approx 303$ m.

3 **[Foundational]** • **Target: Law of Cosines**

In triangle PQR, two sides of length 8 and 11 enclose an angle of $52°$. Find the length of the third side, to two decimal places.

$c^2 = 8^2 + 11^2 - 2(8)(11)\cos 52° \approx 185 - 108.36 = 76.64$, so $c \approx 8.75$.

4 **[Foundational]** • **Target: Law of Cosines**

A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.

The largest angle is opposite the side of length 12: $\cos C = \frac{7^2 + 9^2 - 12^2}{2(7)(9)} = \frac{-14}{126} = -\frac{1}{9}$, so $C \approx 96.4°$.

5 **[Foundational]** • **Target: Law of Cosines**

Find the area of a triangle with sides 10 and 14 enclosing an angle of $35°$, to two decimal places.

$A = \frac{1}{2}(10)(14)\sin 35° \approx 70 \times 0.5736 \approx 40.15$ square units.

6 **[Foundational]** • **Target: Law of Cosines**

A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N\,50°\,E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.

The angle between the legs is $90° + 50° = 140°$. Law of Cosines:
$d^2 = 30^2 + 45^2 - 2(30)(45)\cos 140° \approx 2925 + 2068.4 = 4993.4$, so $d \approx 71$ km.

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7 [Foundational] • Target: The ambiguous case (SSA)

In triangle ABC, $A = 30°$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B.

$\sin B = \frac{b\sin A}{a} = \frac{9(0.5)}{5} = 0.9$. This gives $B \approx 64.2°$ or $B \approx 115.8°$. Since $A + B < 180°$ in both cases, there are two possible triangles.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The ambiguous case (SSA)

In triangle DEF, $D = 40°$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

$\sin E = \frac{e\sin D}{d} = \frac{6\sin 40°}{10} \approx 0.386$, giving $E \approx 22.7°$ or $E \approx 157.3°$. Since $d > e$ (the side opposite the given angle is longer), only the acute solution is valid: exactly one triangle exists.

9 [Intermediate] • Target: Choosing the right law

For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:

a Two angles and one side are known (AAS). Law of Sines — a full angle-side ratio pair is available.

b Three sides are known (SSS). Law of Cosines — no angle is known to pair with a side.

c Two sides and the included angle are known (SAS). Law of Cosines — the included angle lets you find the third side directly.

(a) Law of Sines — a full angle-side ratio pair is available.; (b) Law of Cosines — no angle is known to pair with a side.; (c) Law of Cosines — the included angle lets you find the third side directly.

10 [Intermediate] • Target: Choosing the right law

A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.

Find angle C between the sides of length 120 and 150 (opposite the side 90):
$\cos C = \frac{120^2 + 150^2 - 90^2}{2(120)(150)} = \frac{28800}{36000} = 0.8$, so $C \approx 36.87°$. Area
$= \frac{1}{2}(120)(150)\sin(36.87°) \approx 9000(0.6) = 5400\ \text{m}^2$.

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11 [Intermediate] • Target: Heron's formula

State Heron's formula for the area of a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$, then use it to find the area of a triangle with sides 13, 14, 15.

$A = \sqrt{s(s-a)(s-b)(s-c)}$. Here $s = \frac{13+14+15}{2} = 21$. $A = \sqrt{21(8)(7)(6)} = \sqrt{7056} = 84$.

12 [Intermediate] • Target: Heron's formula

A triangular garden has sides 9 m, 12 m, and 15 m. Use Heron's formula to find its area, and verify it is a right triangle by checking the Pythagorean relationship.

$s = \frac{9+12+15}{2} = 18$. $A = \sqrt{18(9)(6)(3)} = \sqrt{2916} = 54$ m^2. Check: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$ ✓ — confirms a right triangle, and indeed $\frac{1}{2}(9)(12) = 54$ matches.

13 [Intermediate] • Target: Navigation and bearing problems

A plane flies on a bearing of $N\,60°\,E$ for 200 km, then changes course to a bearing of $S\,80°\,E$ for 150 km. Find the angle between the two legs of the flight, and set up (but do not evaluate) the Law of Cosines expression for the total displacement.

The angle between the two bearings: turning from $N60°E$ to $S80°E$ sweeps through $180° - 60° - 80° = 40°$ measured one way, but the interior angle of the path (between the two flight segments) is $180° - 40° = 140°$. Displacement: $d^2 = 200^2 + 150^2 - 2(200)(150)\cos(140°)$.

14 [Intermediate] • Target: Solving oblique triangles from a diagram

In triangle XYZ, $x = 8$, $y = 10$, and angle $Z = 48°$ (between the given sides). Find side z, then find angle X using the Law of Sines, to the nearest tenth of a degree.

$z^2 = 8^2 + 10^2 - 2(8)(10)\cos 48° \approx 164 - 107.0 = 57.0$, so $z \approx 7.55$. Then $\sin X = \frac{8\sin 48°}{7.55} \approx 0.787$, giving $X \approx 51.9°$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying triangle solutions with the angle sum • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

For the triangle solved above ($z \approx 7.55$, $X \approx 51.9°$, $Z = 48°$), find angle Y and verify all three angles sum to $180°$.

$Y = 180° - 48° - 51.9° = 80.1°$. Check: $48° + 51.9° + 80.1° = 180°$ ✓.

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16 **[Challenging]** • **Target: Choosing between multiple valid approaches** • **Cross-Lesson: 7.04 Applications: Growth and Decay**

A triangle has $a = 6$, $b = 8$, $c = 10$. Find angle C (opposite the longest side) two different ways: (1) using the Law of Cosines, and (2) recognizing this as a Pythagorean triple.

(1) $\cos C = \frac{6^2 + 8^2 - 10^2}{2(6)(8)} = \frac{0}{96} = 0 \Rightarrow C = 90°$. (2) Since $6^2 + 8^2 = 36 + 64 = 100 = 10^2$, this is a right triangle by the Pythagorean theorem, confirming $C = 90°$ directly — both methods agree.

17 **[Challenging]** • **Target: Choosing the right law** • **Cross-Lesson: 8.01 Rational Functions and Asymptotes**

In ΔABC, $\angle A = 45^\circ$, $\angle B = 120^\circ$, and side $a = 11$. Find side b (Law of Sines).

$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow b = \frac{11 \sin 120^\circ}{\sin 45^\circ} = \frac{11\sqrt{6}}{2}$. Final Answer: $\frac{11\sqrt{6}}{2}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

18 **[Challenging]** • **Target: Heron's formula** • **Cross-Lesson: 8.02 Graphing Rational Functions**

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side c. Round your answer to two decimal places.

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos C$. Given: $a = 10$, $b = 15$, and $C = 60^\circ$. Substitute the values into the formula: $c^2 = 10^2 + 15^2 - 2(10)(15)\cos 60^\circ$ $c^2 = 100 + 225 - 300 \cdot \frac{1}{2}$ $c^2 = 325 - 150$ $c^2 = 175$ $c = \sqrt{175}$ $c = \sqrt{25 \cdot 7}$ $c = 5\sqrt{7}$ To find the decimal approximation, we calculate: $c \approx 5 \cdot 2.64575$ $c \approx 13.22875$ Rounding to two decimal places.

19 **[Challenging]** • **Target: Navigation and bearing problems** • **Cross-Lesson: 8.03 Introduction to Limits**

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (\sqrt{3})(0) = 2$. $||\mathbf{u}|| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2, \quad ||\mathbf{v}|| = \sqrt{2^2 + 0^2} = 2$. Step 2: Solve for $\cos\theta$: $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{2}{(2)(2)} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \arccos(1/2) = 60^\circ$.

20 **[Challenging]** • **Target: Solving oblique triangles from a diagram** • **Cross-Lesson: 9.01 Arithmetic Sequences and Series**

Find the angle between $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle -1, 1 \rangle$.

$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = -\frac{\sqrt{2}}{2} \Rightarrow \theta = 135^\circ$. Final Answer: 135° Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Law of Sines

State the Law of Sines for any triangle ABC.

a. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ ✓

b. $a^2 = b^2 + c^2 - 2bc\cos A$

c. $a\sin A = b\sin B = c\sin C$

d. $\frac{a}{\cos A} = \frac{b}{\cos B}$

The Law of Sines states that the ratio of each side length to the sine of its opposite angle is equal for all three vertices.

22 [Foundational] • Target: The Law of Sines

In ΔABC, $A = 30^\circ$, $B = 45^\circ$, and $a = 8$. Find the exact length of side b.

a. $8\sqrt{2}$ ✓

b. $4\sqrt{2}$

c. $8\sqrt{3}$

d. 16

$\frac{8}{\sin 30^\circ} = \frac{b}{\sin 45^\circ} \Rightarrow \frac{8}{1/2} = \frac{b}{\sqrt{2}/2} \Rightarrow 16 = \frac{2b}{\sqrt{2}} \Rightarrow b = 8\sqrt{2}$.

23 [Foundational] • Target: The Law of Sines

Under which given information cases can the Law of Sines be directly applied to begin solving a triangle?

a. **AAS or ASA** ✓

b. SAS or SSS

c. SSS only

d. Right triangles only

The Law of Sines requires at least one known angle-side opposite pair, which is always provided in AAS and ASA cases.

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24 [Foundational] • Target: The Ambiguous Case (SSA)

In the ambiguous SSA case with acute angle A, what does $h = b\sin A$ represent?

a. The altitude from vertex C to side c ✓

b. The perimeter of the triangle

c. The hypotenuse

d. The length of side a

$h = b\sin A$ is the perpendicular distance (height) from vertex C to the opposite line.

25 [Foundational] • Target: The Ambiguous Case (SSA)

If $A = 30^\circ$, $b = 10$, and $a = 4$, how many triangles can be formed?

a. 0 (no triangle exists) ✓

b. 1 triangle

c. 2 triangles

d. Infinitely many

$h = 10\sin(30^\circ) = 5$. Since $a = 4 < h = 5$, the side is too short to reach the opposite baseline, so no triangle can be formed.

26 [Foundational] • Target: The Ambiguous Case (SSA)

If $A = 30^\circ$, $b = 10$, and $a = 7$, how many triangles can be formed?

a. 2 triangles ✓

b. 1 triangle

c. 0 triangles

d. Cannot be determined

$h = 10(0.5) = 5$. Since $h < a < b$ $(5 < 7 < 10)$, side a can swing to two different locations, forming exactly 2 triangles.

27 [Foundational] • Target: The Law of Cosines

State the Law of Cosines to solve for side c given sides a, b and included angle C.

a. $c^2 = a^2 + b^2 - 2ab\cos C$ ✓

b. $c^2 = a^2 + b^2 + 2ab\cos C$

c. $c = a\cos B + b\cos A$

d. $c^2 = a^2 + b^2 - ab\sin C$

The Law of Cosines is the generalized Pythagorean theorem: $c^2 = a^2 + b^2 - 2ab\cos C$.

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28 [Foundational] • Target: The Law of Cosines

In ΔABC, $a = 5$, $b = 7$, and $C = 60^\circ$. Find the exact length of side c.

a. $\sqrt{39}$ ✓

b. $\sqrt{74}$

c. $\sqrt{49} = 7$

d. 6

$c^2 = 5^2 + 7^2 - 2(5)(7)\cos(60^\circ) = 25 + 49 - 70(0.5) = 74 - 35 = 39 \Rightarrow c = \sqrt{39}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Law of Cosines

In ΔABC, $a = 3$, $b = 5$, $c = 7$. Find the exact value of $\cos C$.

a. $-1/2$ ✓

b. $1/2$

c. $-1/5$

d. $-3/7$

$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{9 + 25 - 49}{2(3)(5)} = \frac{34 - 49}{30} = \frac{-15}{30} = -\frac{1}{2}$.

30 [Intermediate] • Target: Area of an Oblique Triangle

State the sine formula for the area of ΔABC given sides a and b and angle C.

a. $\text{Area} = \frac{1}{2}ab\sin C$ ✓

b. $\text{Area} = ab\sin C$

c. $\text{Area} = \frac{1}{2}ab\cos C$

d. $\text{Area} = \frac{a+b}{2}\sin C$

Using altitude $h = b\sin C$, base a gives $\text{Area} = \frac{1}{2}ab\sin C$.

31 [Intermediate] • Target: Area of an Oblique Triangle

Find the exact area of ΔABC if $a = 6$, $b = 10$, and $C = 30^\circ$.

a. 15 ✓

b. 30

c. $15\sqrt{3}$

d. 7.5

$\text{Area} = \frac{1}{2}(6)(10)\sin(30^\circ) = 30(1/2) = 15$.

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32 [Intermediate] • Target: Area of an Oblique Triangle

Heron's formula states Area $= \sqrt{s(s-a)(s-b)(s-c)}$. Find the area of a triangle with sides 5, 6, and 7.

a. $6\sqrt{6}$ ✓

b. $12\sqrt{3}$

c. 18

d. 14

Semi-perimeter $s = \frac{5+6+7}{2} = 9$. Area
$= \sqrt{9(9-5)(9-6)(9-7)} = \sqrt{9(4)(3)(2)} = \sqrt{216} = \sqrt{36 \times 6} = 6\sqrt{6}$.

33 [Intermediate] • Target: Area of an Oblique Triangle

In ΔABC, $\angle A = 120^\circ$, $\angle B = 50^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.

a. 9.6

b. 11.6

c. 5.0

d. **10.6** ✓

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{12 \sin 50^\circ}{\sin 120^\circ} \approx 10.6$.

34 [Intermediate] • Target: Law of Sines

In ΔABC, $\angle A = 30^\circ$, $\angle B = 40^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.

a. **12.9** ✓

b. 11.9

c. 13.3

d. 13.9

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{10 \sin 40^\circ}{\sin 30^\circ} \approx 12.9$.

35 [Intermediate] • Target: Law of Cosines

In ΔABC, $\angle A = 45^\circ$, $\angle B = 70^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.

a. 14.3

b. **13.3** ✓

c. 12.3

d. 15.6

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{10 \sin 70^\circ}{\sin 45^\circ} \approx 13.3$.

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36 [Intermediate] • Target: The ambiguous case (SSA)

In ΔABC, $\angle A = 30^\circ$, $\angle B = 50^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.

a. 19.4

b. 20.0

c. **18.4** ✓

d. 17.4

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{12 \sin 50^\circ}{\sin 30^\circ} \approx 18.4$.

37 [Intermediate] • Target: Choosing the right law

In ΔABC, $\angle A = 30^\circ$, $\angle B = 80^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.

a. 22.6

b. 32.0

c. **23.6** ✓

d. 24.6

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{12 \sin 80^\circ}{\sin 30^\circ} \approx 23.6$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.02 Graphing Rational Functions

In ΔABC, $\angle A = 60^\circ$, $\angle B = 50^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.

a. 7.8

b. 9.8

c. 8.3

d. **8.8** ✓

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{10 \sin 50^\circ}{\sin 60^\circ} \approx 8.8$.

39 [Challenging] • Target: Navigation and bearing problems • Cross-Lesson: 8.03 Introduction to Limits

In ΔABC, $\angle A = 120^\circ$, $\angle B = 40^\circ$, and side $a = 8$. Find side b (to one decimal place) using the Law of Sines.

a. 6.9

b. 2.7

c. **5.9** ✓

d. 4.9

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{8 \sin 40^\circ}{\sin 120^\circ} \approx 5.9$.

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40 [Challenging] • Target: Solving oblique triangles from a diagram • Cross-Lesson: 9.01 Arithmetic Sequences and Series

In ΔABC, $\angle A = 75^\circ$, $\angle B = 50^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.

a. 8.0

b. **9.5** ✓

c. 8.5

d. 10.5

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{12 \sin 50^\circ}{\sin 75^\circ} \approx 9.5$.

41 [Challenging] • Target: Verifying triangle solutions with the angle sum • Cross-Lesson: 9.02 Geometric Sequences and Series

In ΔABC, $\angle A = 75^\circ$, $\angle B = 40^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.

a. 9.0

b. **8.0** ✓

c. 7.0

d. 6.4

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{12 \sin 40^\circ}{\sin 75^\circ} \approx 8.0$.

42 [Challenging] • Target: Choosing between multiple valid approaches • Cross-Lesson: 9.03 The Binomial Theorem

In ΔABC, $\angle A = 60^\circ$, $\angle B = 40^\circ$, and side $a = 12$. Find side b (to one decimal place) using the Law of Sines.

a. 9.9

b. **8.9** ✓

c. 7.9

d. 8.0

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{12 \sin 40^\circ}{\sin 60^\circ} \approx 8.9$.

43 [Challenging] • Target: The Law of Sines • Cross-Lesson: 1.01 Functions and Function Notation

In ΔABC, $\angle A = 75^\circ$, $\angle B = 70^\circ$, and side $a = 10$. Find side b (to one decimal place) using the Law of Sines.

a. 10.7

b. 8.7

c. **9.7** ✓

d. 9.3

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{10 \sin 70^\circ}{\sin 75^\circ} \approx 9.7$.

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44 [Challenging] • Target: The Ambiguous Case (SSA) • Cross-Lesson: 1.02 Domain and Range

In ΔABC, $\angle A = 45^\circ$, $\angle B = 80^\circ$, and side $a = 8$. Find side b (to one decimal place) using the Law of Sines.

a. 10.1

b. 12.1

c. 14.2

d. **11.1** ✓

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{8\sin 80^\circ}{\sin 45^\circ} \approx 11.1$.

45 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 1.03 Average Rate of Change

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.

a. **5.7** ✓

b. 6.7

c. 7.7

d. 4.7

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{5\sin 80^\circ}{\sin 60^\circ} \approx 5.7$.

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